Zhexi Luo 羅喆熙

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Research Interests

Dexterous manipulation and grasping; robot foundation models; learning from human demonstrations.
Keywords: Dexterity; generalization.

Education

Sun Yat-sen University — **Guangzhou, China**
Bachelor of Engineering in Computer Science and Technology — *Sep 2022–Jun 2026*
School of Computer Science and Engineering
Average score: 85/100

Research Highlights

∗ Equal contribution.

DexMimic: A Generalist Dexterous Hand-Object Tracking Policy via Value-Guided Reset Allocation
Zhexi Luo, Zhixuan Xu, Rui Yang, Zeyu Xiao, Yiwen Hou, Lin Shao.
Under review, Sep 2026. Paper Project

OmniDexGrasp: Generalizable Dexterous Grasping via Foundation Model and Force Feedback
Yi-Lin Wei*, **Zhexi Luo***, Yuhao Lin, Mu Lin, Zhizhao Liang, Shuoyu Chen, Wei-Shi Zheng.
IEEE International Conference on Robotics and Automation (ICRA), 2026. Paper Project

Collaborative Projects

T(R,O) Grasp++: Cross-Embodiment Dexterous Grasping in Cluttered Environments at Scale
Xin Fei*, Zhixuan Xu*, Huaicong Fang, Fangjun Chen, Zhaoyu Mu, Rui Yang, **Zhexi Luo**, Lin Shao.
Conference on Robot Learning (CoRL), 2026. Paper Project

Humanoid Whole-Body Manipulation via Active Spatial Brain and Generalizable Action Cerebellum
Zhizhao Liang*, Yi-Lin Wei*, Xuhang Chen*, Mu Lin, Yi-Xiang He, **Zhexi Luo**, Jun-Hui Liu, Kun-Yu Lin, Wei-Shi Zheng.
European Conference on Computer Vision (ECCV), 2026 (Oral). Paper Project

AXIS: A Growable Community-Driven Data Engine for Scalable Robot Manipulation
Mengfei Zhao*, Dihong Huang*, Yikai Tang*, Peihao Li*, Mingxuan Yan*, Ruiqi Zhuang*, Yanjia Huang*, Jie Wang, Hai Zhai, Tony Zhou, Rui Zhang, **Zhexi Luo**, Yuchen Huang, Jianfei Yang, Jiachen Li.
Physical World Models for Scaling Embodied AI @ IROS 2026. Technical Report Website

DriftTrace: Combating Concept Drift in Security Applications Through Detection and Explanation
Yuedong Pan, Lixin Zhao, Tao Leng, **Zhexi Luo**, Lijun Cai, Aimin Yu, Dan Meng.
IEEE Transactions on Information Forensics and Security (TIFS), vol. 21, pp. 1957–1972, 2026. Paper

Research Experience

LinS Lab, National University of Singapore (NUS) — **Singapore**
Research Intern, Supervisor: Prof. Lin Shao — *Nov 2025–Present*
Projects: DexMimic; T(R,O) Grasp++.
- Train generalist dexterous manipulation policies through large-scale reinforcement learning.
- Harness generative foundation models to synthesize dexterous grasp data at scale.

Axis Robotics — **Silicon Valley, CA, USA**
Researcher — *Jan–May 2026*
Projects: AXIS.
- Built scalable web-based infrastructure for robot data collection.

Intelligence SciencE and systEm Lab (iSEE), Sun Yat-sen University — **Guangzhou, China**
Undergraduate Researcher, Supervisor: Prof. Wei-Shi Zheng — *Apr–Sep 2025*
Projects: OmniDexGrasp; Humanoid Whole-Body Manipulation.
- Developed generalizable dexterous grasping across prompts, hands, and tasks using foundation models.
- Developed humanoid whole-body manipulation using agents and a generalist controller.

Technical Skills

Programming: Python, C, C++, JavaScript, TypeScript.
Tools: Blender, Viser, LaTeX.
Simulation: Isaac Gym, MuJoCo.
Hardware: Wuji and Inspire hands; Kinova and xArm manipulators.

Honors and Academic Service

2023: National First Prize, China Undergraduate Mathematical Contest in Modeling.

Conference Reviewer: CVPR 2026, ECCV 2026, IROS 2026, NeurIPS 2026, ICRA 2027.